

Figure 8.6 Rickets and Osteomalacia

Toxicity:

Vitamin D toxicity can occur if one consumes over 10,000 IU per day (1000 IU per day in children) for several months. This causes hypervitaminosis D, which increases the blood calcium levels. This results in bone loss and kidney stones. Long-term overconsumption of vitamin D can cause calcification of organs such as the heart, blood vessels and the kidneys.

8.1.3 Vitamin E:

Vitamin E comprises of a group of isoprenoid substituted compounds called tocopherols (Fig. 8.7). It also includes tocotrienols. The most common form of vitamin E is alpha tocopherol which is present in large quantities in human body. The most abundant form of vitamin E in diet is gamma tocopherol.

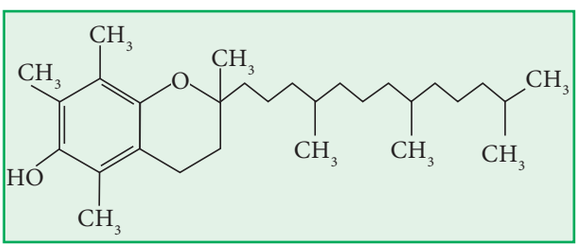


Figure. 8.7 Structure of vitamin E (alpha tocopherol)

Sources:

Cotton seed oil, sun flower oil, wheat germ oil and leafy vegetables are the rich natural sources (Figure. 8.8).

Groundnut oil: 261mg/100g

Wheat germ oil: 150 mg/100g

Unit 8.indd 203 21-05-2018 15:11:39